



SERANGOON JUNIOR COLLEGE
2009 JC2 PRELIMINARY EXAMINATIONS

MATHEMATICS

Higher 2

9740/Paper 1

Monday

19 August 2009

Additional materials: Writing paper

List of Formulae (MF15)

TIME : 3 hours

READ THESE INSTRUCTIONS FIRST

Write your name and class on the cover page and on all the work you hand in.

Write in dark or black pen on both sides of the paper.

You may use a soft pencil for any diagrams or graphs.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** the questions.

Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.

You are expected to use a graphic calculator.

Unsupported answers from a graphic calculator are allowed unless a question specifically states otherwise.

Where unsupported answers from a graphic calculator are not allowed in a question, you are required to present the mathematical steps using mathematical notations and not calculator commands.

You are reminded of the need for clear presentation in your answers.

The number of marks is given in brackets [] at the end of each question or part question.

At the end of the examination, fasten all your work securely together.

This question paper consists of 8 printed pages.

Answer all questions (100 marks)

1. Solve $|2 - 3x| > 2 + 3x - x^2$. [4]

2(a) With respect to an origin O , the points A , B and C have position vectors $2\mathbf{i} + 4\mathbf{j} + \mathbf{k}$, $14\mathbf{i} + \mathbf{k}$ and $2\mathbf{i} + 5\mathbf{j} + \mathbf{k}$ respectively. The point P on AB is such that $AP : PB = 1 : \lambda$. Given that $\overline{CP}$ is parallel to the vector $2\mathbf{i} + 4\mathbf{j} + \mathbf{k}$, find λ . [3]

(b) The angle between the vector $\mathbf{i} + \mu\mathbf{k}$ and the vector $\mathbf{j} + 2\mathbf{k}$ is 150° . Find the exact value of the constant μ . [3]

3 Show that the equation $y = \frac{4x^2 + 14x + 11}{2(x+1)}$ can be written as $y = 2x + A + \frac{B}{2(x+1)}$ where A and B are constants to be determined. [2]

Hence, state the sequence of transformations which transform the graph of $y = x + \frac{1}{x}$ to the graph of $y = \frac{4x^2 + 14x + 11}{2(x+1)}$. [4]

4 Prove by induction that

$$(2^2)(1) + (3^2)(2) + (4^2)(2^2) + (5^2)(2^3) + \dots + (n+1)^2(2^{n-1}) = (n^2 + 2)2^n - 2$$

for all positive integers n . [5]

- 5 Three students, Sally (**S**), Preeti (**P**) and Chunli (**C**) have been tasked to build 80 models for sale during their college's fundraising campaign. Three steps **J**, **K** and **L** are required in the building of each model. Due to the different proficiency levels of the students, the time that they take to complete each step is different. The following table provides the breakdown of time that each student takes to complete each step in seconds:

	J (sec)	K (sec)	L(sec)
S	210	30	10
P	100	15	7
C	350	40	12

The total time taken by the students to complete the 80 models is 21240 seconds while the total time taken for the students to complete step **L** is 790 seconds. Find the number of models that each student builds. [4]

- 6 The functions f and g are defined as follows:

$$f: x \rightarrow 3 - 2e^{x+1}, \quad x \in \mathbb{R}$$

$$g: x \rightarrow x^2 + 2x + 2, \quad x \in \mathbb{R}, x \leq -1.$$

- (a) Find g^{-1} , giving its domain. [3]
- (b) Explain why the composite function fg exists. Find fg in similar form and state its exact range. [3]

[Turn over]

7 The curve C has equation

$$y = \frac{x^2 + b}{x + a}, \text{ where } a > 0 \text{ and } b > 0.$$

- (i) State the coordinates of the intersection(s) of the curve C with the axes in terms of a and b . [1]
- (ii) Find the equation(s) of the asymptote(s). [2]
- (iii) Draw a sketch of the curve C , labeling the equation(s) of its asymptote(s) and coordinates of any intersection with the axes. [2]
- (iv) Hence find the range of values of k , where k is a positive constant, for which the equation $x^2 + b = (x + a)(kx - a)$ has no real root. [2]

8 Sketch on an Argand diagram, the set of points representing all complex numbers z such that

$$|z + 1 - i| \leq |1 + i| \text{ and } |z| \geq |z - 2i|. \quad [4]$$

Hence find the range of $|z|$ and $\arg(z)$. [3]

9(a) Use the substitution $x = 2 \tan \theta$ to find the exact value of $\int_{\frac{2}{\sqrt{3}}}^2 \frac{1}{x(\sqrt{x^2 + 4})} dx$. [6]

(b) (i) By writing $x = A(1 - x) + B$, where A and B are constants,

prove that

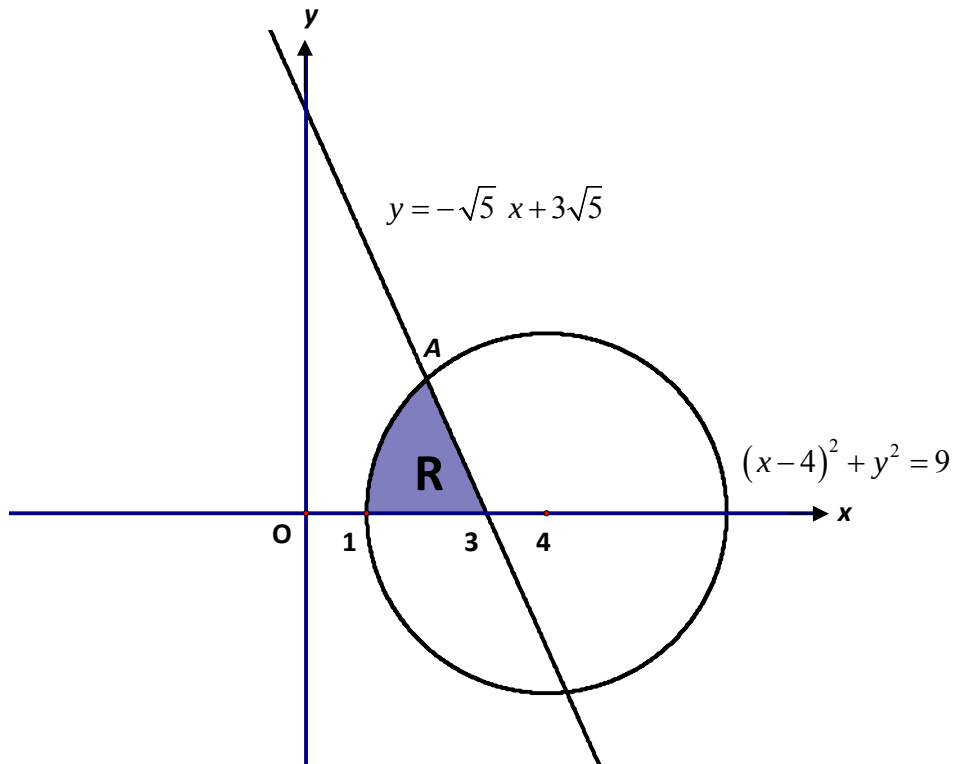
$$\int \frac{x}{\sqrt{1-x}} dx = \frac{2}{3}(1-x)^{\frac{3}{2}} - 2(1-x)^{\frac{1}{2}} + C$$

where C is an arbitrary constant.

[2]

(ii) Hence, find $\int \sqrt{x} (\sin^{-1} \sqrt{x}) dx$. [3]

- 10 The diagram shows the region **R** bounded by the circle with equation $(x-4)^2 + y^2 = 9$, the straight line $y = -\sqrt{5}x + 3\sqrt{5}$ and the x -axis. The coordinates of **A**, which is the intersection point of the line $y = -\sqrt{5}x + 3\sqrt{5}$ and the circle, is $(2, \sqrt{5})$.



- (a) Find the area of the region **R**, giving your answer correct to 3 decimal places. [3]
- (b) Show that the volume, S , formed when **R** is rotated through 2π radians about the y -axis is

$$S = \pi \int_0^{\sqrt{5}} \left[\frac{(y - 3\sqrt{5})^2}{5} - (25 - y^2 - 8\sqrt{9 - y^2}) \right] dy. \quad [4]$$

[There is no need to evaluate S .]

[Turn over]

- 11(i) Given that $y = \ln\left(\frac{\cos x}{e-x}\right)$, show that $\frac{dy}{dx} = -\tan x + \frac{1}{(e-x)}$. By further differentiation of this result, or otherwise, find the Maclaurin's series of y up to and including the term in x^3 . [6]
- (ii) Deduce the Maclaurin's series for the equation $y = \ln\left(\frac{\cos(ex)}{1+x}\right)$ up to and including the term in x^3 . [2]
- 12(a) The variables x and y are related by the differential equation $x \frac{dy}{dx} - y + \frac{2}{x} = 0$. Show that by means of the substitution $y = ux$, the differential equation may be reduced to $\frac{du}{dx} = -\frac{2}{x^3}$. [2]
- Hence, or otherwise, find the general solution for y in terms of x . [2]
- Sketch, for $x > 0$, two members of the family of solution curves corresponding to a positive value and a negative value of the arbitrary constant. [2]
- (b) The rate at which a chemical evaporates at room temperature is proportional to the amount of substance which has not yet evaporated. If the initial amount of chemical was 500g and the amount which has evaporated at time t is x . Write down a differential equation involving x and t . Show that $x = 500 - Ae^{-kt}$, where A and k are constants. [2]
- Show that the time taken for half the chemical to evaporate is $\frac{1}{k} \ln 2$. [3]

- 13 It is given that p , q and v are the first three terms of a geometric progression. It is also given that p , v and q are the first, third and fourth term of an arithmetic progression respectively.

(i) Show that

$$q^2 = pv \quad \text{and} \quad v = \frac{2q + p}{3} \quad [2]$$

(ii) Hence show that $3\left(\frac{q}{p}\right)^2 - 2\left(\frac{q}{p}\right) - 1 = 0$. [2]

(iii) Find the sum to infinity in terms of p for the geometric progression. [2]

14(a) Given $f(r) = \frac{1}{r^2}$, $r \neq 1$, show that

$$f(r-1) - f(r) = \frac{2r-1}{r^2(r-1)^2}. \quad [1]$$

(i) Hence or otherwise, find the sum to infinity of the series $\frac{3}{2^2(1)^2} + \frac{5}{3^2(2)^2} + \frac{7}{4^2(3)^2} + \dots$ [3]

(ii) Deduce that $\sum_{r=2}^n \frac{2r-1}{r^2(r+1)^2} < 1$. [2]

[Turn over]

(b)

The sequence of positive numbers $x_1, x_2, x_3, \dots$ satisfies the recurrence

relation $x_{n+1} = 4 - \frac{5}{x_n + 2}$ for $n \geq 1$.

- (i) As $n \rightarrow \infty$, $x_n \rightarrow \alpha$, find the exact value of α . [2]
- (ii) Show that if $x_n > \alpha$, then $x_{n+1} < x_n$. [2]
- (iii) Use a calculator to determine the behavior of the sequence $\{x_n\}$ when $x_1 = 4$. Hence state briefly how the results in (ii) relate to the behavior of the sequence. [2]